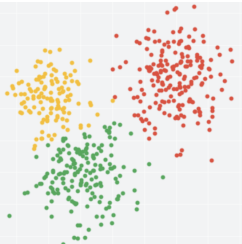


K-Means Algorithm

Procheta Sen



Representative-based algorithms

- Let k be the number of clusters
- Let $D = \{\bar{X}_1, \dots, \bar{X}_n\}$ be the dataset
- The goal is to determine k representatives $\bar{Y}_1, \dots, \bar{Y}_k$ that minimise the following objective function

$$\sum_{i=1}^n \left[\min_{1 \leq j \leq k} d(\bar{X}_i, \bar{Y}_j) \right]$$

i.e. the sum of the distances of the objects to their closest representatives needs to be minimised.

Representative-based algorithms

We obtain specific algorithms by specifying

- the way of choosing representatives, and
- the distance function $d(\cdot, \cdot)$

In general representatives do not necessarily belong to the dataset.

General k -representatives approach

- **Initialise:** pick initial k representatives
- **Iteratively refine:**
 - ▶ **(assign step)** Assign each object to its closest representative using distance function $d(\cdot, \cdot)$. Denote the corresponding clusters $C_1, \dots, C_k$.
 - ▶ **(optimise step)** Determine the optimal representative $\bar{Y}_j$ for each cluster C_j that minimises its local objective function

$$\sum_{\bar{X}_i \in C_j} d(\bar{X}_i, \bar{Y}_j)$$

k -Means algorithm

- ▶ Representatives are chosen **not necessarily** from the dataset
- ▶ The distance function is **squared Euclidean distance**

k -Means algorithm

The objective function

$$\min_{\bar{Y}_1, \dots, \bar{Y}_k} \sum_{i=1}^k \sum_{\bar{X} \in C_i} \|\bar{X} - \bar{Y}_i\|^2$$

where C_i consists of the objects that are closest to $\bar{Y}_i$.

We want to minimise the total **squared Euclidean distance** between data objects and their cluster representatives $\bar{Y}_1, \dots, \bar{Y}_k$

This objective function is called the *within cluster sum of squares (WCSS)* objective

k -Means algorithm

Assume that the clusters $C_1, C_2, \dots, C_k$ are fixed.

Find the set $\bar{Y}_1, \bar{Y}_2, \dots, \bar{Y}_k$ of representatives such that

$$f_{C_1, \dots, C_k}(\bar{Y}_1, \dots, \bar{Y}_k) = \sum_{i=1}^k \sum_{\bar{X} \in C_i} \|\bar{X} - \bar{Y}_i\|^2$$

is minimised.

$$\frac{\partial f_{C_1, \dots, C_k}(\bar{Y}_1, \dots, \bar{Y}_k)}{\partial \bar{Y}_i} = - \sum_{\bar{X} \in C_i} 2(\bar{X} - \bar{Y}_i) = 0$$
$$\bar{Y}_i = \frac{1}{|C_i|} \sum_{\bar{X} \in C_i} \bar{X}$$

Just compute the centroid (mean) of each cluster and that will give you the new **optimal** cluster representatives

KMeans Illustration

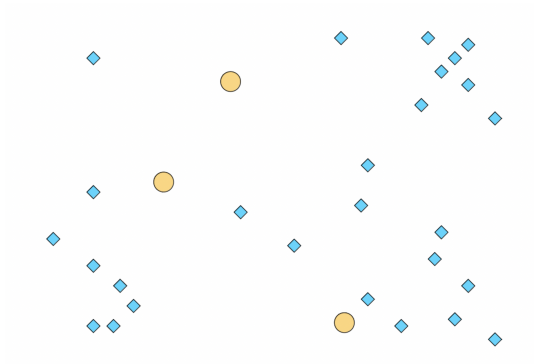


Figure: K Means Demonstration

KMeans Illustration

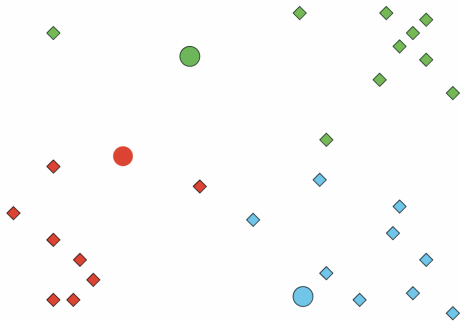


Figure: K Means Demonstration

KMeans Illustration

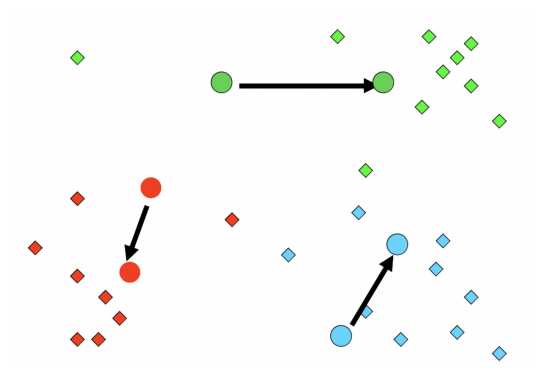


Figure: K Means Demonstration

KMeans Illustration

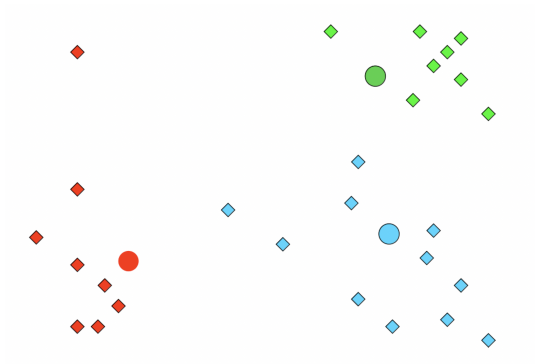


Figure: K Means Demonstration

Issues with k -Means algorithm

- ▶ Results can vary depending on initial random choices
- ▶ Can get trapped in a local minimum that isn't the global optimal solution
- ▶ Repeat the clustering procedure multiple times with different initialisations and select the *best* final clustering
- ▶ Outliers have a larger effect on the mean value, hence cluster centre and the cluster
- ▶ Cluster centres (means) are not actual instances in the cluster
- ▶ Euclidean distance used in the algorithm is inappropriate for categorical features